

Design, Model, and Test Advanced Communication and Sensing Electronics at Cryogenic Temperatures

Sponsor: WSU Systems-on-Chip Lab and Boeing

Mentor: Subhanshu Gupta

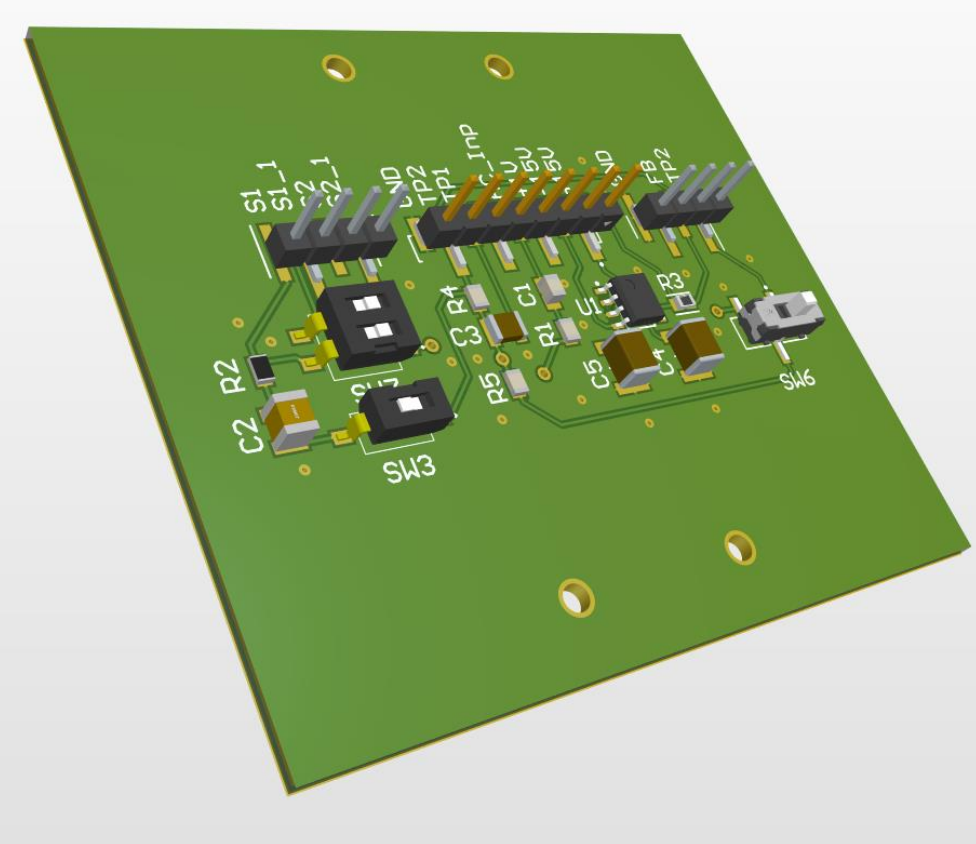
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Cryogenic Chamber / Component Testing



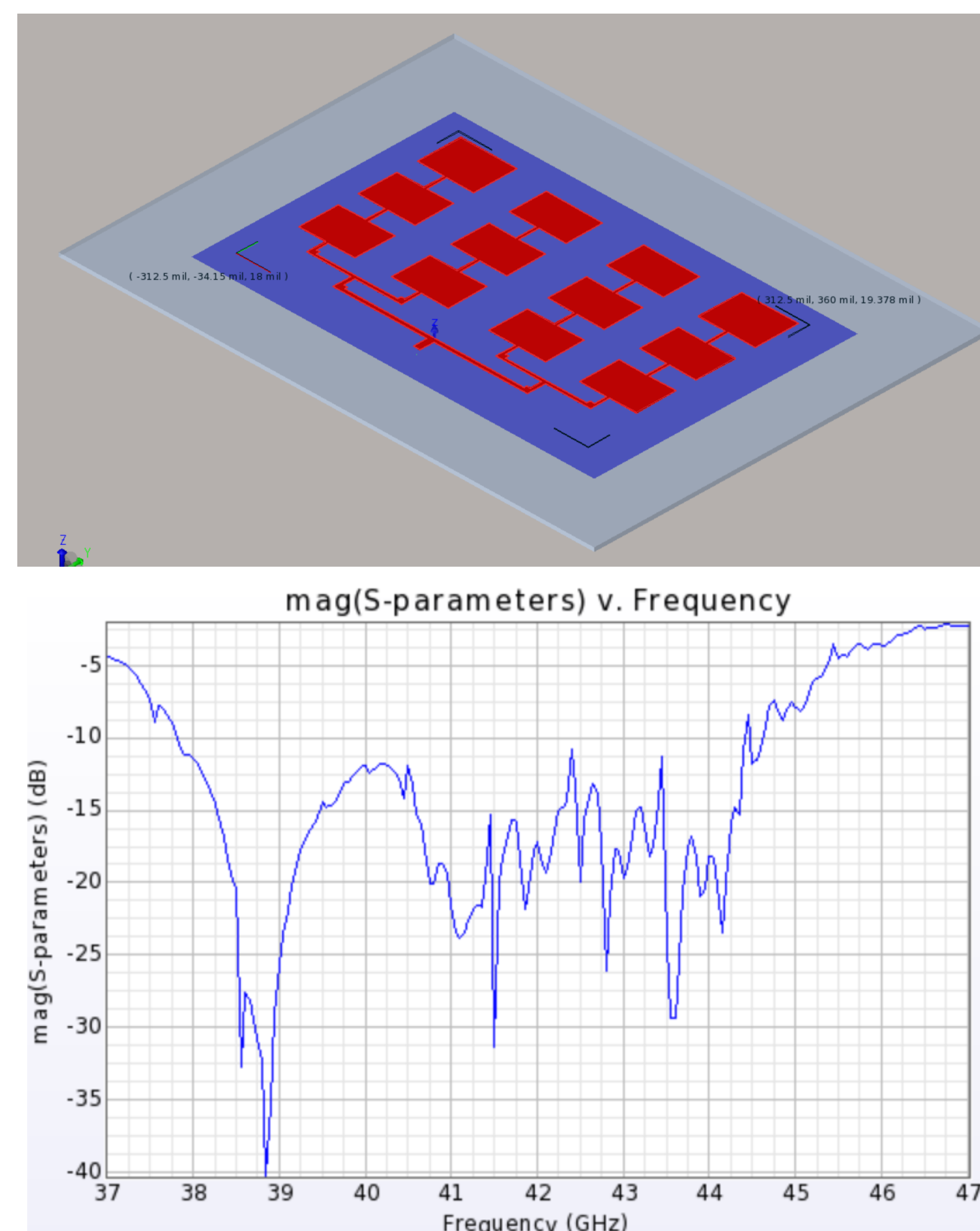
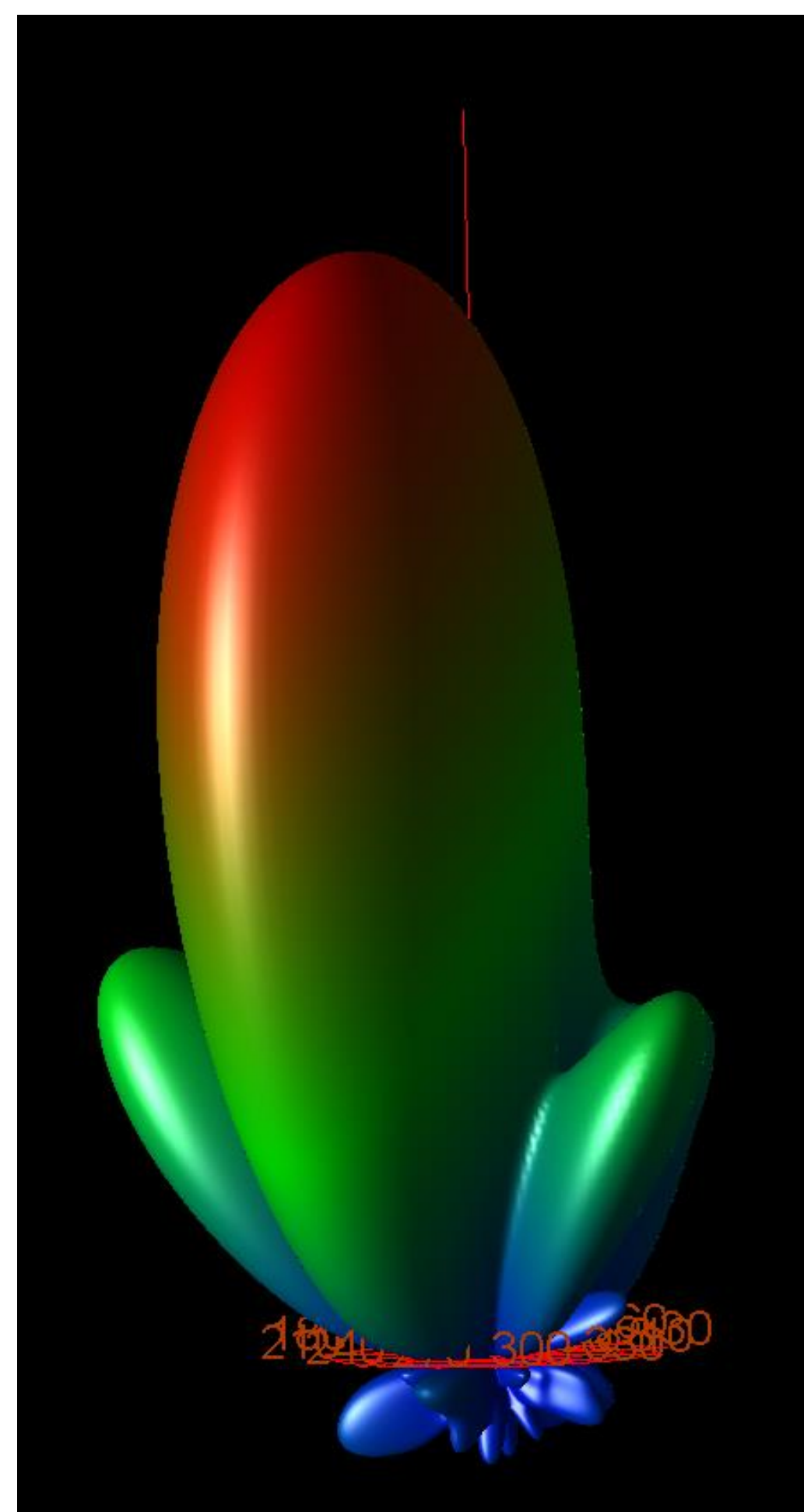
Cryo-cooling lowers temperatures to reduce thermal noise, improving electronic performance and reliability.

- The HyPress ICE-T cooler drops electronics from room temperature to 4K for cryo testing.
- Characterized components at cryogenic temperatures using custom designed test boards.
- Tested low noise operational amplifiers, Gallium Nitride-based FETs, voltage references, various capacitor technologies and many more.



Antenna Design

Designed a 3×4 patch antenna array for wide bandwidth (40–44 GHz) on 8-mil Rogers 4003C in ADS, achieving return loss below -10 dB. Final design shows 13.2 dBi directivity and 12.24 dBi gain, ideal for short-range radar.

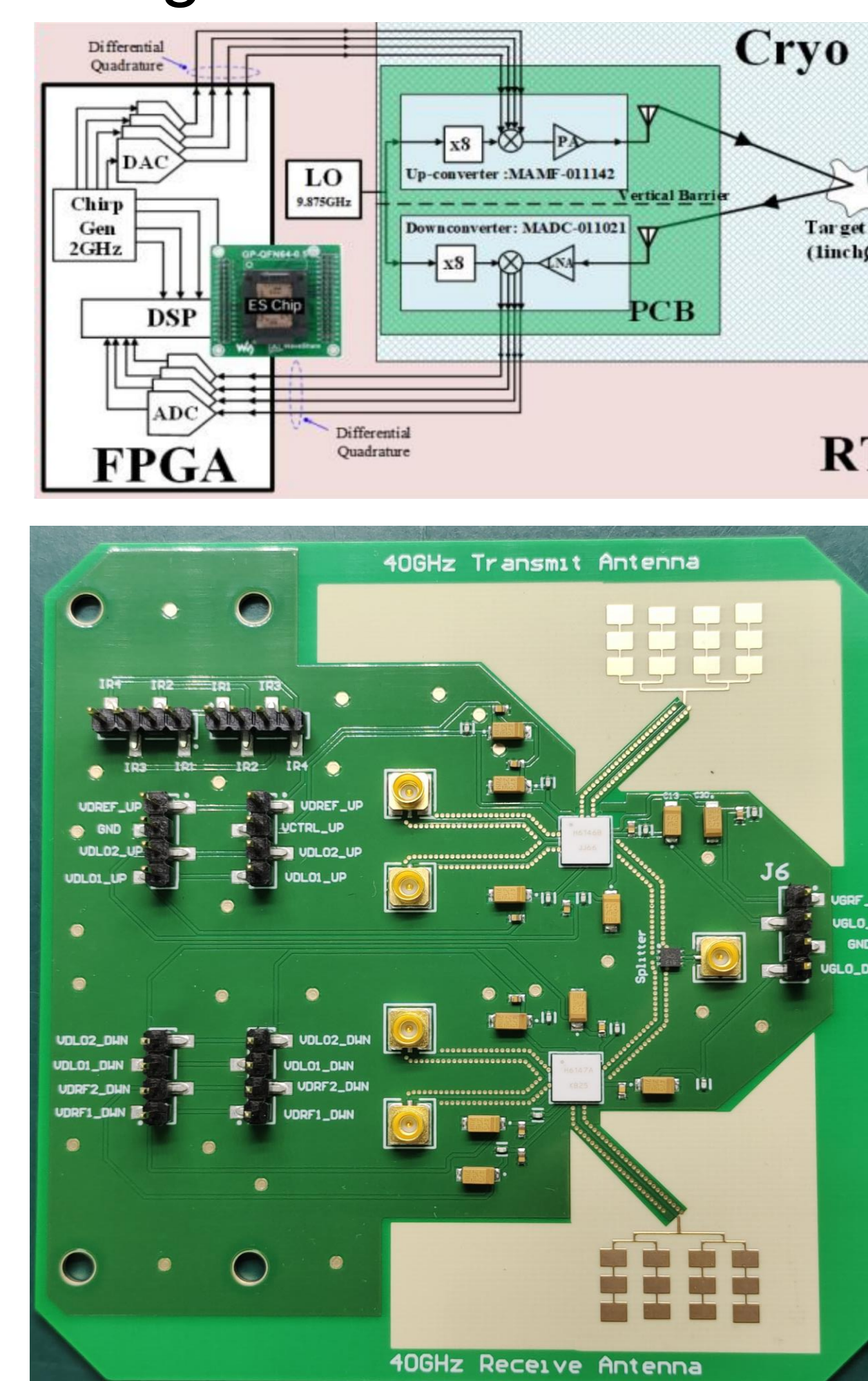


Radar Concept/Objective

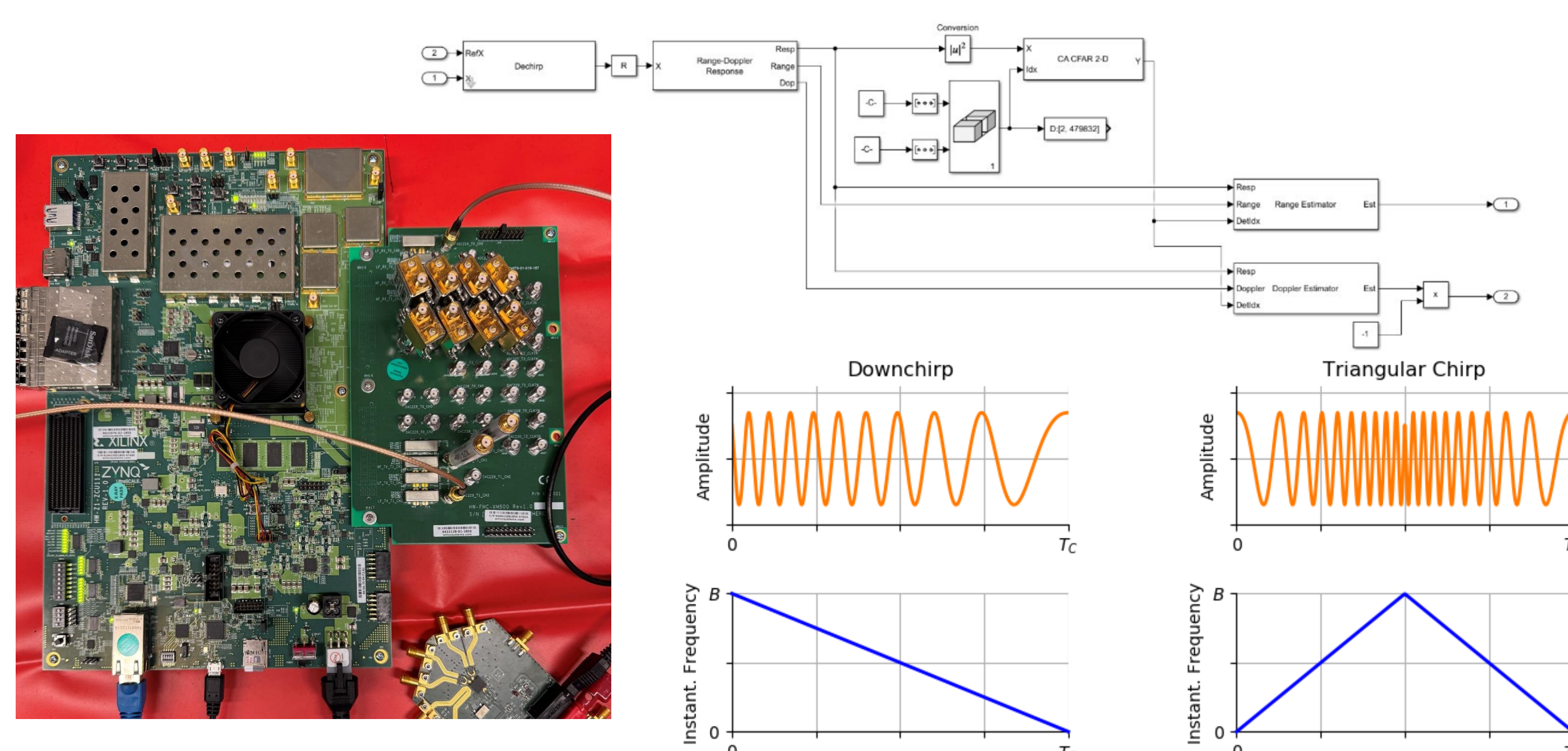
The RADAR design uses a FMCW signal to allow for sensitive range and velocity estimation of targets, while utilizing lower power than pulse variants. Cryo-cooling of the transmitter and receiver front end is theorized to increase detection probability through reduction in thermal noise of electronics on board.

Board Design

A compact 4-layer RF board was designed in Altium for the 40–44 GHz Radar, with internal ground planes supporting the antenna, integrated up/down conversion paths, grounded coplanar waveguides, and an empty backside for thermal contact with the chamber.



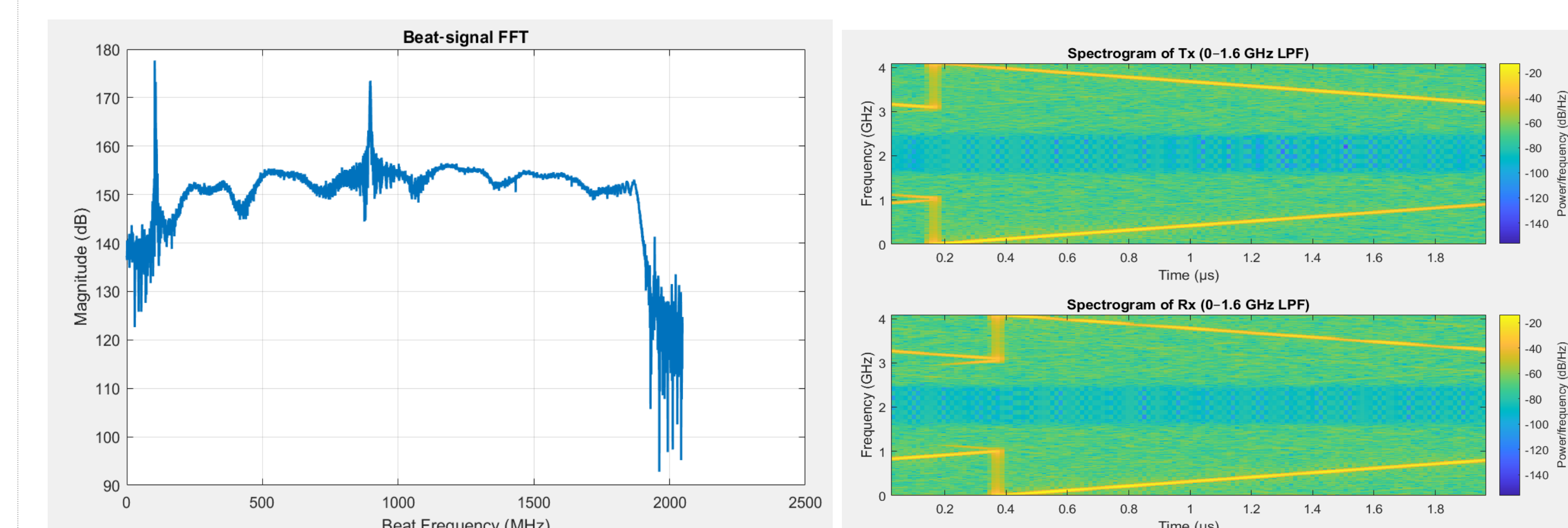
Signal Generation / DSP



A commercial RFSoc (ZCU111) provides RF capable DACs and ADCs for 4 GHz bandwidth chirp signal generation and return signal sampling for receiver processing. Sampled data is mixed with chirp signal and processed in MATLAB/Simulink for frequency content, which corresponds to threshold target detection, distance, and velocity.

Future Work

The next steps for this project will focus on the cryogenic testing of the radar board to evaluate its performance in extremely low temperatures. These tests will also close knowledge gaps and confirm the boards suitability for cryogenic applications, paving the way for further design improvements in advanced sensing electronics and space systems. Improvements to bridge the gap between our design and commercially used radars is the inclusion of additional transmit and receive antennas for use in beamforming and angle of arrival estimation.



Applications / Impact

- Automotive mmWave Radar
- Radio Telescopes
- Satellite/Deep Space Communication
- Medical Imaging

Glossary:

ADS: Advanced Design System

dBi: Decibels relative to isotropic

DSP: Digital Signal Processing

FMCW: Frequency Modulated Continuous Wave

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Team Name: **Domat**